SKILLS CRITICAL THINKING

9

4 For electrolysis of each of the following

i write the ionic half-equation for the reaction occurring at the cathode

ii write the ionic half-equation for the reaction occurring at the anode

iii state what has been oxidised and what has been reduced.

a Molten lead(II) bromide using carbon electrodes

b Sodium chloride solution using carbon electrodes

c Calcium bromide solution using carbon electrodes

d Copper(II) sulfate solution using platinum electrodes

e Aluminium nitrate solution using carbon electrodes

f Molten magnesium iodide using carbon electrodes

g Dilute hydrochloric acid using platinum electrodes

SKILLS INTERPRETATION, REASONING

8

5 You are asked to find out whether two compounds, S and T, are electrolytes or non-electrolytes. S melts at 1261 °C and is soluble in water. T melts at 265 °C and is insoluble in water. Describe, with the aid of diagrams, how you would find out if each of these substances was an electrolyte or a non-electrolyte. In each case say what you would look for to help you to decide.

6 When copper(II) sulfate solution is electrolysed using copper electrodes the reaction at the cathode is the same as with inert electrodes but no oxygen is given off at the cathode. Instead the anode gets smaller as copper ions go into solution. The half-equation for the reaction at the anode is:

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

a Write the half-equation for the reaction at the cathode.

SKILLS REASONING

b When copper(II) sulfate solution is electrolysed using inert electrodes the blue colour of the solution fades and the solution becomes more acidic.

i Explain these observations.

9

ii Predict and explain what happens to the colour and acidity of the solution when copper(II) sulfate solution is electrolysed with copper electrodes.

END OF CHEMISTRY ONLY

UNIT QUESTIONS

You may need to refer to the Periodic Table on page 320.

SKILLS CRITICAL THINKING (4) **1** Hydrogen is the most common element in the universe.

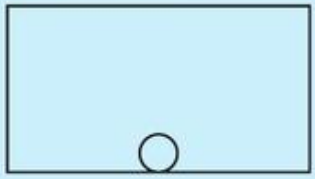
The melting point and boiling point of hydrogen are shown in the table:

Melting point/°C	–259
Boiling point/°C	–253

a Put a cross in the box to show a temperature at which hydrogen is a liquid. **(1)**

–265 °C ☐ –260 °C ☐ –255 °C ☐ –250 °C ☐

SKILLS INTERPRETATION (6) **b** The circle in the diagram represents 1 molecule of hydrogen. Complete the diagram to show the arrangement of particles in liquid hydrogen. You should add at least 10 more circles to the diagram. **(2)**



SKILLS PROBLEM SOLVING (8) **c** Hydrogen has three isotopes, the least common of which is tritium, ^{3}H. Use the Periodic Table to work out the number of protons, neutrons and electrons in an atom of tritium. **(3)**

Number of protons:

Number of neutrons:

Number of electrons:

d Hydrogen reacts with nitrogen to form ammonia (NH_3).

(7) **i** The electronic configuration of a nitrogen atom is (put a cross in one box): **(1)**

2, 2, 3 ☐ 2, 5 ☐ 2, 8, 4 ☐ 2, 2, 8, 2 ☐

SKILLS CRITICAL THINKING (8) **ii** Draw a dot-and-cross diagram, showing outer electrons only, to show the bonding in an ammonia molecule. **(2)**

(6) **e** Ammonia forms salts that contain the ammonium ion. Put a cross in the box to show which of the following formulae for ammonium compounds is incorrect.

NH_4SO_4 ☐ NH_4NO_3 ☐ NH_4Cl ☐ $(NH_4)_2CO_3$ ☐

5

f Two more compounds that contain hydrogen are water and ethanol. Put a cross in the box to show a method that can be used to separate a mixture of ethanol and water. **(1)**

crystallisation	☐	simple distillation	☐
filtration	☐	fractional distillation	☐

(Total 10 marks)

SKILLS CRITICAL THINKING 6

2

a Complete the following passage by using the words below. You may use the words once, more than once, or not at all. **(4)**

mass number	atomic number	groups	periods
protons	electrons	outer shell	nucleus

The elements in the Periodic Table are arranged in order of _____.

The vertical columns are called _____ and contain elements which have the same number of _____ in their _____.

5

b Put a cross next to the symbol(s) of elements that are non-metals. **(2)**

H ☐ V ☐ Pb ☐ Ar ☐ W ☐

SKILLS REASONING 6

c Silicon is an element that shows some of the properties of metals and non-metals. It is sometimes called a *metalloid*.

i Two properties of silicon are:

- silicon conducts electricity
- silicon forms an oxide (SiO_2) that reacts with hot concentrated sodium hydroxide.

Explain which of these is not a property of metals. **(2)**

SKILLS ANALYSIS, REASONING 8

ii The structure of silicon dioxide (SiO_2) is shown in the diagram. This structure continues in all directions.



Explain in terms of electronic configurations and bonding whether the grey or red circles in this diagram represent silicon atoms. **(3)**

iii By referring to the diagram in part ii explain whether you would expect SiO_2 to be a solid, liquid or gas at room temperature. **(2)**

(Total 13 marks)

SKILLS INTERPRETATION, REASONING 7

3 Strontium is an element in Group 2 of the Periodic Table and bromine is an element in Group 7.

a Explain what happens in terms of electrons when an atom of strontium combines with an atom of bromine to form strontium bromide. You may draw a diagram to illustrate your answer. **(3)**

b Explain in terms of structure and bonding whether you would expect strontium bromide to have a high or a low melting point. **(3)**

c Explain what you understand by the term 'relative atomic mass' of an element. **(2)**

SKILLS PROBLEM SOLVING 10

d The natural abundances of the two isotopes of bromine are:

^{79}Br 50.69%

^{81}Br 49.31%

Calculate the relative atomic mass of bromine. Give your answer to 2 decimal places. **(2)**

SKILLS CRITICAL THINKING 6

e The formula of strontium nitrate is (put a cross in one box): **(1)**

$SrNO_3$ ☐ $Sr(NO_3)_2$ ☐ $SrNO_{32}$ ☐ Sr_3N_2 ☐

(Total 11 marks)

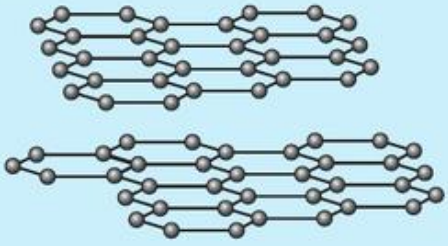
SKILLS REASONING 9

4 The diagram represents the structure of diamond, a form of carbon.

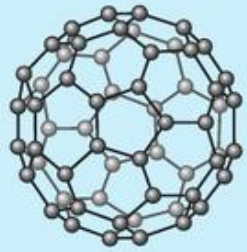


a Explain in terms of structure and bonding why diamond is very hard. **(2)**

b Two more forms of carbon are graphite and C_{60} fullerene. Part of the structure of graphite and a molecule of C_{60} fullerene are shown in the diagram.



graphite



C_{60} fullerene

i Explain whether graphite or C_{60} fullerene has the higher boiling point. **(4)**

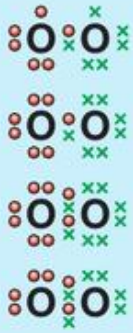
ii Explain why graphite conducts electricity. **(2)**

iii Explain why C_{60} fullerene does not conduct electricity. **(2)**

(Total 10 marks)

SKILLS ANALYSIS 7

5 **a** Oxygen gas has the formula O_2. Which of the following is a correct dot-and-cross diagram showing the bonding in O_2 (outer shells of electrons only are shown). Circle one of the diagrams. **(1)**



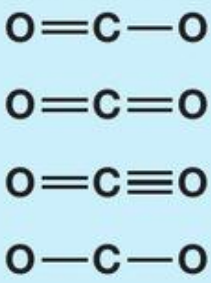
SKILLS CRITICAL THINKING 6

b Oxygen reacts with potassium to form potassium oxide. Put a cross in one box to indicate what the formula of potassium oxide is. **(1)**

PO ☐ KO ☐ K_2O ☐ KO_2 ☐

SKILLS ANALYSIS

c Oxygen reacts with carbon to form carbon dioxide. Which diagram shows the covalent bonds in carbon dioxide? Circle one diagram. **(1)**



(Total 3 marks)

SKILLS PROBLEM SOLVING 7

6 In an experiment to find the empirical formula of lead oxide, a small porcelain dish was weighed, filled with lead oxide and weighed again. The dish was placed in a tube and heated in a stream of hydrogen. The hydrogen reduced the lead oxide to a bead of metallic lead. When the apparatus was cool, the dish and its bead of lead were weighed together.

Mass of porcelain dish = 17.95 g

Mass of porcelain dish + lead oxide = 24.80 g

Mass of porcelain dish + lead = 24.16 g

(A_r: O = 16, Pb = 207)

a Calculate the mass of lead in the lead oxide. **(1)**

b Calculate the mass of oxygen in the lead oxide. **(1)**

9

c There are three different oxides of lead: PbO, PbO_2 and Pb_3O_4. Use your results from **a** and **b** to find the empirical formula of the oxide used in the experiment. **(3)**

(Total 5 marks)

SKILLS PROBLEM SOLVING 7

7 Copper reacts with concentrated nitric acid to produce copper(II) nitrate solution and nitrogen dioxide gas.

$$Cu(s) + 4HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l)$$

A student carried out an experiment to make some copper(II) nitrate crystals.

a The student started with 2.00 g of copper and added excess nitric acid. Calculate the maximum mass of copper(II) nitrate, $Cu(NO_3)_2$, which could be obtained. (A_r: N = 14, O = 16, Cu = 63.5) **(3)**

SKILLS EXECUTIVE FUNCTION 5

b Explain the method the student could use to obtain crystals of copper(II) nitrate from the copper nitrate solution. **(3)**

SKILLS PROBLEM SOLVING 7

c The copper nitrate crystallises out of the solution as $Cu(NO_3)_2 \cdot 3H_2O$. The student did some calculations and worked out that he should make 7.61 g of crystals. He actually only made 5.23 g. Calculate the percentage yield of the student's experiment. **(2)**

(Total 8 marks)

CHEMISTRY ONLY

SKILLS PROBLEM SOLVING 9

8 If pyrite (FeS_2) is heated strongly in air it reacts according to the equation:

$$4FeS_2(s) + 11O_2(g) \rightarrow 2Fe_2O_3(s) + 8SO_2(g)$$

Iron can be extracted from the iron(III) oxide produced, and the sulfur dioxide can be converted into sulfuric acid. In this question you can assume that the molar volume of any gas at rtp is 24 dm^3.

a Calculate the mass of iron(III) oxide that can be obtained from 480 kg of pure pyrite. (A_r: O = 16, S = 32, Fe = 56) **(3)**

b Calculate the volume of sulfur dioxide (measured at rtp) produced from 480 kg of pyrite. **(2)**

c The next stage of the manufacture of sulfuric acid is to convert the sulfur dioxide into sulfur trioxide (SO_3) by reacting it with oxygen.

i Write a balanced equation for this reaction. **(2)**

ii Calculate the volume of oxygen (measured at rtp) needed for the complete conversion of the sulfur dioxide produced in **b** into sulfur trioxide. **(2)**

(Total 9 marks)

SKILLS CRITICAL THINKING 7

9 Strontium hydroxide, $Sr(OH)_2$, is only sparingly soluble in water at room temperature. In an experiment to measure its solubility, a student made a saturated solution of strontium hydroxide. She pipetted 25.0 cm^3 of this solution into a conical flask, added a few drops of methyl orange indicator and then titrated it with 0.100 mol/dm^3 hydrochloric acid from a burette. She needed to add 32.8 cm^3 of the acid to neutralise the strontium hydroxide.

$$Sr(OH)_2(aq) + 2HCl(aq) \rightarrow SrCl_2(aq) + 2H_2O(l)$$

5

a Explain what is meant by a *saturated solution*. **(1)**

SKILLS PROBLEM SOLVING 8

b Calculate the number of moles of HCl in 32.8 cm^3 of 0.100 mol/dm^3 hydrochloric acid. **(1)**

c Calculate the concentration of the strontium hydroxide in mol/dm^3. **(3)**

d The solubility of a substance is usually measured in units of g per 100 g of water. The mass of 1 dm^3 of water is 1000 g. Use the value that you obtained in **c** to calculate the solubility of strontium hydroxide as g of strontium hydroxide per 100 g of water. (A_r: H = 1, O = 16, Sr = 88) **(2)**

SKILLS EXECUTIVE FUNCTION 7

e The calculation used in **d** gives only an approximate value for the solubility because the volume changes when a substance is dissolved in water. Describe an experiment that the student could carry out to determine a more accurate value for how much strontium hydroxide dissolves in 100 g of water at this temperature. You can assume that you are given 50 cm^3 of saturated strontium hydroxide solution. You should include a description of the apparatus you would use and the measurements you would make. **(5)**

(Total 12 marks)

SKILLS PROBLEM SOLVING 8 **10** A student carried out some experiments using dilute sulfuric acid.

a He found that 25.0 cm^3 of 0.100 mol/dm^3 sodium hydroxide solution was neutralised by 20.0 cm^3 of the dilute sulfuric acid.

The equation for the reaction is

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Calculate the concentration of the sulfuric acid in mol/dm^3. **(4)**

9

b 100 cm^3 of this same sulfuric acid was reacted with magnesium.

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

i Magnesium is a metal. Describe the structure and bonding in a piece of magnesium. **(3)**

ii Magnesium is malleable. Explain in terms of structure and bonding why magnesium is malleable. **(2)**

iii Calculate the amount in moles of sulfuric acid that was used. **(1)**

iv The student used 0.100 g of magnesium in the experiment. Work out whether the sulfuric acid or the magnesium was in excess. **(2)**

v Calculate the volume of hydrogen gas that would be produced at rtp in this experiment. (Assume that the molar volume at rtp of hydrogen is 24 000 cm^3.) **(2)**

vi When the student did the experiment they collected 94 cm^3 of gas. Calculate the percentage yield of the experiment. **(1)**

c The student electrolysed the solution of magnesium sulfate.

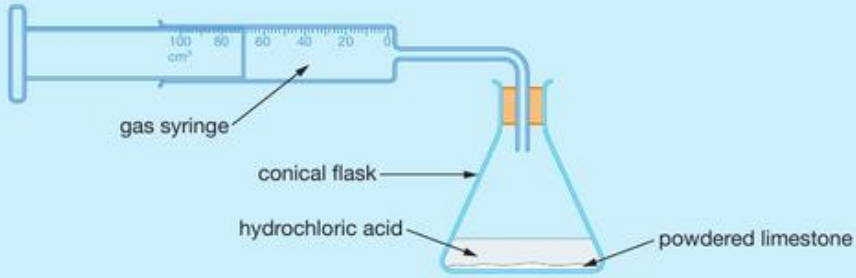
i State the name of the product formed at the cathode. **(1)**

ii Oxygen gas is formed at the anode. Write an ionic half-equation for the formation of oxygen at the anode. **(2)**

(Total 18 marks)

SKILLS CRITICAL THINKING 3

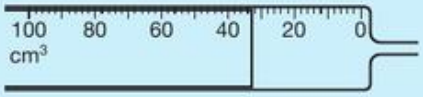
11 A class of 11 students carried out an experiment to determine how much calcium carbonate ($CaCO_3$) was in a sample of limestone. Each student was given a different mass of limestone, which they added to 50 cm^3 of dilute hydrochloric acid and then measured the volume of carbon dioxide produced using the apparatus shown below.



a Name a piece of apparatus that could be used to measure out 50 cm^3 of dilute hydrochloric acid. **(1)**

4

b The diagram shows the reading from the experiment carried out by Student 2.

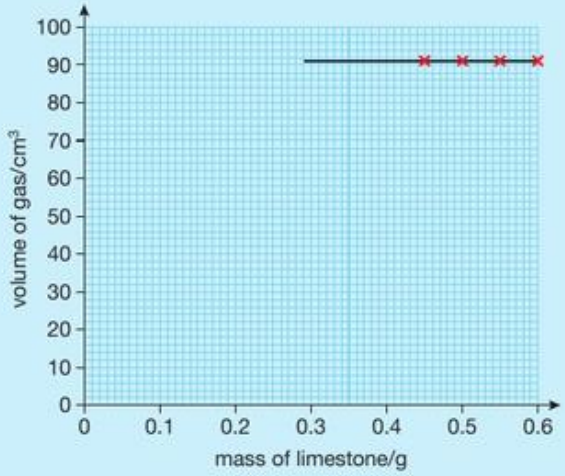


Complete the table with the volume of gas collected by Student 2. **(1)**

Student	Mass of limestone/g	Volume of gas produced/cm^3
1	0.10	22
2	0.15	
3	0.20	37
4	0.25	56
5	0.30	66
6	0.35	78
7	0.40	88
8	0.45	91
9	0.50	91
10	0.55	91
11	0.60	91

6

c Some of the data have been plotted on the graph below. Plot the rest of the data on the graph and draw another straight line of best fit. **(3)**



d The teacher realised that one of the student's results was incorrect.

SKILLS ANALYSIS 4

i Explain which student's result is incorrect. **(2)**

SKILLS DECISION MAKING

ii Suggest one thing that could have gone wrong with the student's experiment to produce this result. **(1)**

SKILLS REASONING 6

e Explain why the volume of gas produced in the experiments carried out by Students 8–11 is the same. **(2)**

SKILLS ANALYSIS

f Use the graph to determine the mass of limestone that reacts exactly with the hydrochloric acid. **(1)**

SKILLS PROBLEM SOLVING

g The equation for the reaction between calcium carbonate and hydrochloric acid is

$$CaCO_3(s) + 2HCl(\ldots) \rightarrow CaCl_2(aq) + H_2O(\ldots) + CO_2(g)$$

7

i Complete the equation by adding the missing state symbols. **(2)**

9

ii Calculate the mass of calcium carbonate that would produce 91 cm^3 of carbon dioxide at rtp. **(3)**

(Assume that the molar volume of a gas at rtp is 24 000 cm^3.)

iii Calculate the percentage of calcium carbonate in the sample of limestone. **(2)**

(Total 18 marks)

END OF CHEMISTRY ONLY

THE ALKALI METALS 123 | THE HALOGENS 130 | GASES IN THE ATMOSPHERE 137 | REACTIVITY SERIES 145
EXTRACTION AND USES OF METALS 160 | ACIDS, ALKALIS AND TITRATIONS 167
ACIDS, BASES AND SALT PREPARATIONS 173 | CHEMICAL TESTS 190

UNIT 2
INORGANIC CHEMISTRY

Inorganic chemistry is the study of all the elements in the Periodic Table and the compounds they form, except organic compounds formed by carbon. The properties of these elements are very different and they form a huge variety of compounds. The Periodic Table is the great unifying principle used by inorganic chemists as a guide to understanding the behaviour of the elements and their compounds. Most of the elements in the Periodic Table are metals and these are some of the most important materials that we use in everyday life. However, scientists are always searching for new materials with exciting properties. Inorganic chemists are involved in the development of these new materials, for instance high-temperature superconductors that are used in trains that levitate above the tracks.



▲ Figure 11.1 The maglev train levitates above the track due to superconducting materials.